

Exercises – 60mins

TITLE: UNDERSTANDING BIG-O NOTATION & ALGORITHM APPLICATIONS

SUBTITLE: CLASSROOM EXERCISES & REAL-WORLD SCENARIOS

INSTRUCTOR: NJOROGE KINYANJUI

Class-based Exercises

Objectives

- Understand Big-O notation
- Analyze algorithm efficiency
- Identify suitable data structures and algorithms
- Apply concepts to real-world problems

Class-based Exercises

Match the Complexity

Instructions: Match each algorithm to its Big-O complexity.

Algorithm	Complexity
a) Binary Search	()
b) Linear Search	()
c) Bubble Sort	()
d) Constant Time Access	()
e) Merge Sort	()

Options:

1. $O(1)$
2. $O(n)$
3. $O(n \log n)$
4. $O(n^2)$
5. $O(\log n)$

Class-based Exercises

Code Analysis

Instructions: Determine the Big-O time complexity of the following functions.

```
# Function 1
def example1(arr):
    for i in arr:
        print(i)

# Function 2
def example2(arr):
    for i in arr:
        for j in arr:
            print(i, j)

# Function 3
def example3(n):
    while n > 1:
        n = n // 2
```

Class-based Exercises

Matching Real-World Applications

Instructions: Match each real-world problem to the most appropriate data structure.

Real-world scenario	Data Structure
a) Auto-complete in Google search	()
b) Undo operation in MS Word	()
c) Shortest route in Google Maps	()
d) Browser history	()
e) Call Center Queue	()

Options:

1. Stack
2. Graph Algorithm
3. Queue
4. Hash Table
5. Binary Search Tree

Class-based Exercises

Identifying Algorithm types

Instructions: Match each approach with its algorithmic type.

Scenario	Algorithm Type
a) Trying all password combinations	()
b) Solving a maze using all paths	()
c) Binary Search	()
d) Merge Sort	()
e) Random quicksort pivot	()

Options:

1. Brute Force
2. Backtracking
3. Divide and Conquer
4. Randomized
5. Greedy

Class-based Exercises

Design Challenge

Prompt: You are designing a school management system. **Identify:**

1. Data structures for:

1. Student records
2. Class scheduling
3. Exam grading

2. Algorithms for:

1. Searching a student
2. Ranking students
3. Allocating classrooms

Activity: Discuss in groups and present your design and justification.

Class-based Exercises

Memory Flash – Flash Round

- What's the complexity of accessing a hash table? ($O(1)$)
- What algorithm uses divide and recursion? (Merge Sort)
- Which structure follows LIFO? (Stack)
- What is the best search method for a sorted list? (Binary Search)
- Which algorithm grows the fastest? (Exponential / $O(2^n)$)